

immgenT Explorer (Seurat Shiny App)

An interactive Shiny app for exploring immgenT Seurat objects with hierarchical filters (choose your experiment > batch (IGT) > sample exp/IGT/sample), FeaturePlot & DimPlot (with adjustable dot size), and a metadata summary table for the loaded samples. You can upload your own `.rds` / `.RData` Seurat object or point the app at a local folder of demo objects.

Image: `immgen_app1:0.1` **Port:** 3838 (HTTP) **CPU/GPU:** CPU only **Architecture:** built for `linux/amd64/`

1) What's inside

In the UI you can choose:

- **Upload a file** — upload a Seurat `.rds` / `.RData`.
- **Use bundled immgenT**

The seurat object requires:

- **RNA counts**
 - a reduction named as: `mde_incremental`
 - sample name in the column `sample_code`
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2) Easiest way (no terminal): Docker Desktop

1. Install Docker Desktop

- Download Docker Desktop [<https://www.docker.com/products/docker-desktop/>]
- Install it
- Open it (!) and go through the default setup (login can be skipped) once until it says “Docker engine running”.

2. Recommended: Increase memory/CPU for bigger objects

- Docker Desktop → **Settings** → **Resources**:
 - **Memory:** 8–16 GB (more for very large datasets)
 - **CPUs:** 4+ cores
 - Click **Apply & Restart**

3. Run the app

- Open the folder `immgenT_App1`
- Windows: Click `open_windows.bat`
- Mac:
 - Click `open_mac.command`
 - The app won't open until you authorize it: Go to your computer's **Settings** → **Security settings**

- The Terminal will open, the code will run. It takes a minute so be patient.
- Your browser should open to **http://localhost:3838**

4. In the app

- Choose **Upload a file** (if you'll upload), or
- Choose **Use bundled demo**

5. Start/Stop later

- Close the terminal

3) Performance & memory tips

- In Docker Desktop → **Settings** → **Resources**, increase **Memory** (8–16 GB+) and **CPUs** (4+) if you work with large Seurat objects.
- Close other containers/apps to free RAM.
- If a dataset is huge, try subsetting/downsampling before loading.

4) About

- **App:** immgenT Explorer (Seurat Shiny)
- **Maintainer:** David Zemmour (dzemmour)
- **Tag:** 0.1
- **License:** MIT license